

Regression Analysis of Airbnb Listing Prices Based on Listing Attributes

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Stat 311

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INTRODUCTION

This analysis aims to investigate the factors that influence AirBnB listing prices. Specifically, the primary research question is:

Which listing characteristics, host attributes, and availability factors significantly affect the price of an AirBnB listing?

Predictors such as property attributes, listing characteristics, host features and availability have potential to influence the price in systematic ways. Understanding these relationships is important from both sides of the transaction (customers and hosts). Hosts can use information similar to what may be found in this analysis to determine optimal pricing strategies, improve listing competitiveness and maximize revenue. At the same time, customers can better understand what properties may be properly priced (good vs bad value) and what factors contribute most to price, helping make informed decisions.

To complete the analysis, exploratory analysis is performed to understand the structure of the data, identify patterns, and detect some potential early flags that could lead to issues such as skewness or outliers. An initial MLR model is then fitted using price to evaluate relationships between predictors and price. Analysis of the initial model will be performed to assess whether regression assumptions are satisfied. Based on findings, transformations or alternative modeling strategies will be considered. The final model will be selected based on statistical adequacy, interpretability and simplicity.

DATASET

The dataset used in this analysis was compiled by InsideAirbnb.com, a platform focused on providing publicly accessible Airbnb data and analysis. The data is collected through web scraping techniques applied to Airbnb listings. This particular dataset includes listings located in Denver, Colorado. It contains several thousand observations, providing a sufficiently large sample size for regression analysis. The dataset includes a wide range of variables related to pricing, property characteristics, host information, review metrics, and availability.

Each observation in the dataset represents a single Airbnb listing created by a host. Listing prices are determined by hosts based on various factors, including property size, amenities, location, and perceived demand. Additional variables, such as review counts and ratings, are generated through guest interactions with the platform. Because the dataset is observational in nature and reflects real-world market behavior, the relationships identified in this analysis represent associations rather than causal effects. The dataset also exhibits characteristics typical of real-world data, including variability in property attributes, skewness in price, and the presence of extreme values.

The variables used for this analysis are as follows

Dependent Variables (DV, Response):

- **Price** - Listed in Dollars (\$), Numeric
- **Log(Price)** - Logarithm of price (Used to address skewness and improve model fit, this will be addressed later), Numeric/transform

Independent Variables (IV, Predictors):

Property Characteristics (Numeric)

x_1 = **Accommodates** - Number of guests the listing can accomodate

x_2 = **Bathrooms** - Number of Bathrooms in the listing

x_3 = **Extra_beds**¹ - Difference in number of beds and bedrooms in a listing

x_4 = **Bedrooms** - Number of bedrooms in the listing

Reviews (numerical)

x_5 = **Reviews_per_month** - Average number of reviews per month on a listing

x_6 = **Review_scores_rating** - Overall review score given by guests

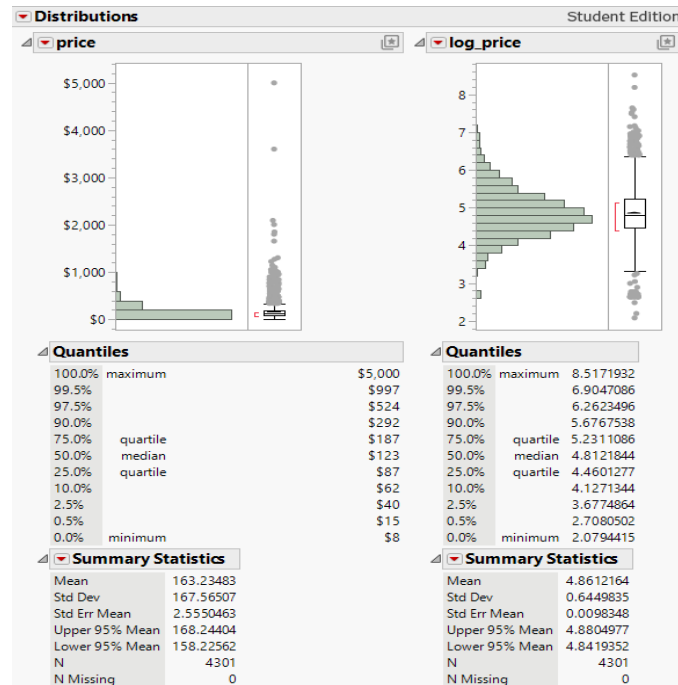
Listing Attributes (categorical) – Using Dummy Variables

x_{7-9} = **Room_type** - Type of listing (entire home/apt, private room, shared room, hotel room)

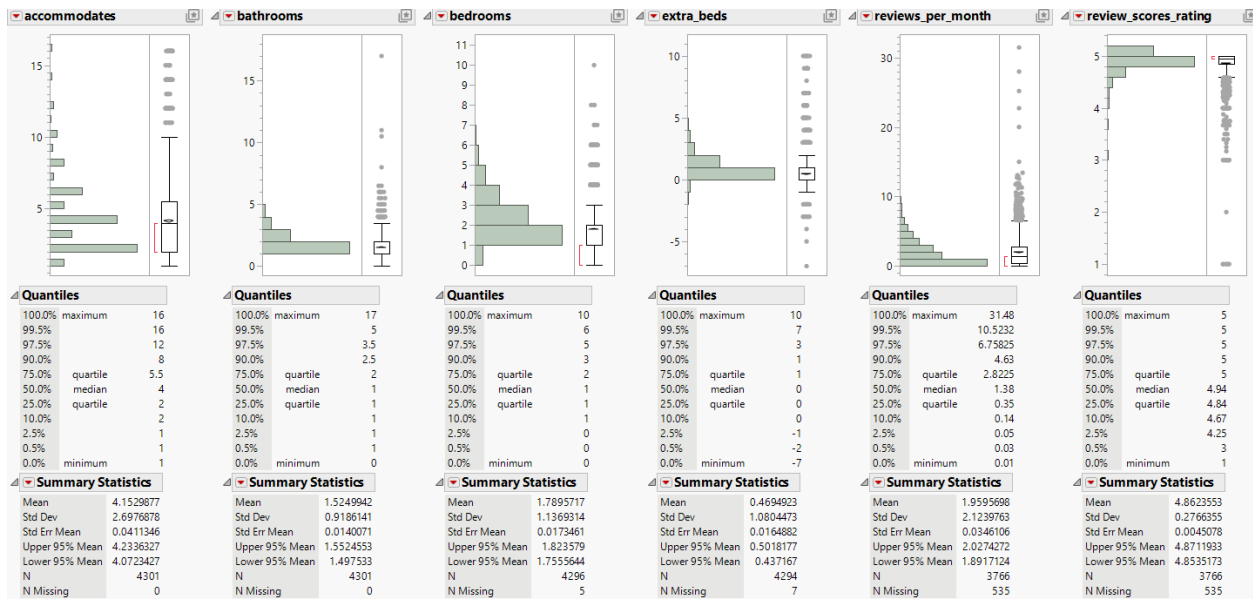
Baseline = entire home/apt

$$x_7 = \begin{cases} 1 & \text{if private room} \\ 0 & \text{if otherwise} \end{cases} \quad x_8 = \begin{cases} 1 & \text{if shared room} \\ 0 & \text{if otherwise} \end{cases} \quad x_9 = \begin{cases} 1 & \text{if hotel room} \\ 0 & \text{if otherwise} \end{cases}$$

EXPLORATORY DATA ANALYSIS (EDA)



The distribution of price is heavily right-skewed, with most listings concentrated at lower values and a long tail of higher-priced listings, including some extreme values up to \$5,000. This is also reflected in the summary statistics, where the mean (\$163) is noticeably higher than the median (\$123), confirming the skewness. In contrast, the distribution of log(price) is much more symmetric and centered, with the mean and median much closer together. The log transformation compresses the higher values and reduces the impact of extreme listings, resulting in a more balanced spread of the data. Because of this, log(price) is more appropriate for regression, as it better satisfies key assumptions such as normality of the errors and constant variance.



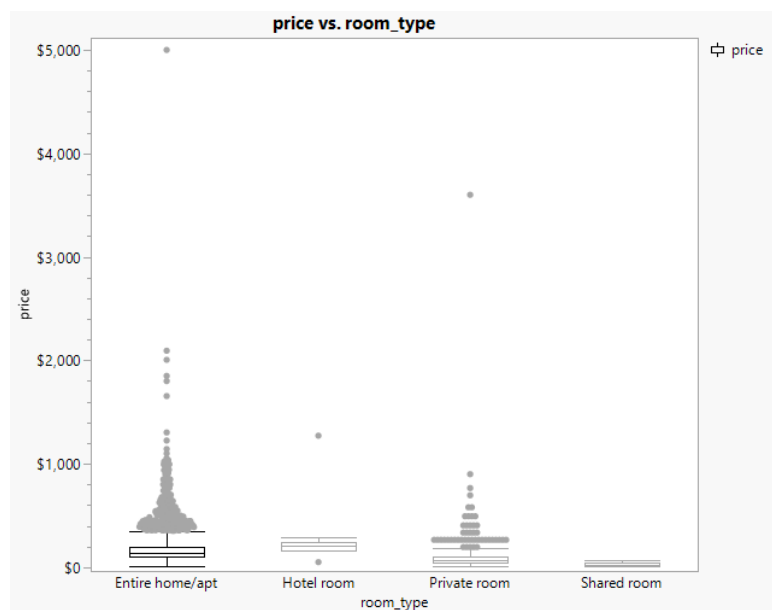
The distributions of the predictor variables show that most size-related variables, such as **accommodates**, **bedrooms**, and **bathrooms**, are reasonably centered with slight right skew and a few higher-value outliers. The **extra_beds** variable is mostly concentrated around zero, indicating that many listings follow a one-bed-per-bedroom setup, with some variation capturing shared sleeping arrangements.

Reviews_per_month is heavily right-skewed, with most listings having low review activity and a few with much higher values. **Review_scores_rating** is tightly clustered near the upper end of the scale, showing that most listings receive high ratings with limited variation. Overall, the variables appear usable for regression, though some

skewness and outliers are present.



The relationship between price and the size-related variables shows a general upward trend, although it is somewhat distorted by outliers. In general, listings that accommodate more guests and have more bathrooms and bedrooms tend to have higher prices. Interestingly, the `extra_beds` variable shows more of a loose bell-shaped pattern, suggesting that sleeping arrangements, not just the total number of beds, may play a role in affecting price.



Price varies by room_type with entire home/apartment listings generally having higher prices and shared rooms with the lowest. Somewhat expected if you were to venture a guess and rent on your own.

Overall, the EDA suggests that while several predictors show relationships with price, the presence of skewness, outliers, and variability indicates that model refinement and transformation may be necessary.

METHODS AND ANALYSIS

This study uses an observational design based on the dataset described above. Because data is not collected through a controlled experiment, the relationships identified show associations rather than causal effects. The following models use multiple linear regression to model price as a function of property characteristics, listing attributes and review-related variables. Higher-order and interaction terms were considered, but exploratory analysis did not suggest strong nonlinear relationships, and the focus was placed on maintaining a simple and interpretable model. Entries with missing price values were excluded. Outliers noticed in the EDA were included as they appeared to be legitimate entries with no evidence suggesting they were invalid². Variance Inflation Factors (VIF) were examined to assess multicollinearity among predictors. Moderate multicollinearity was observed between accommodates and bedrooms, which is expected due to their related interpretation as measures of listing size. However, all variables were retained in the model. Influential observations were also considered using Cook's Distance. While some observations exhibited higher influence, they were determined to be valid data points and were retained in the analysis.

Model 1:

$$E\{y\} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + \beta_8 x_8 + \beta_9 x_9$$

Hypothesis:

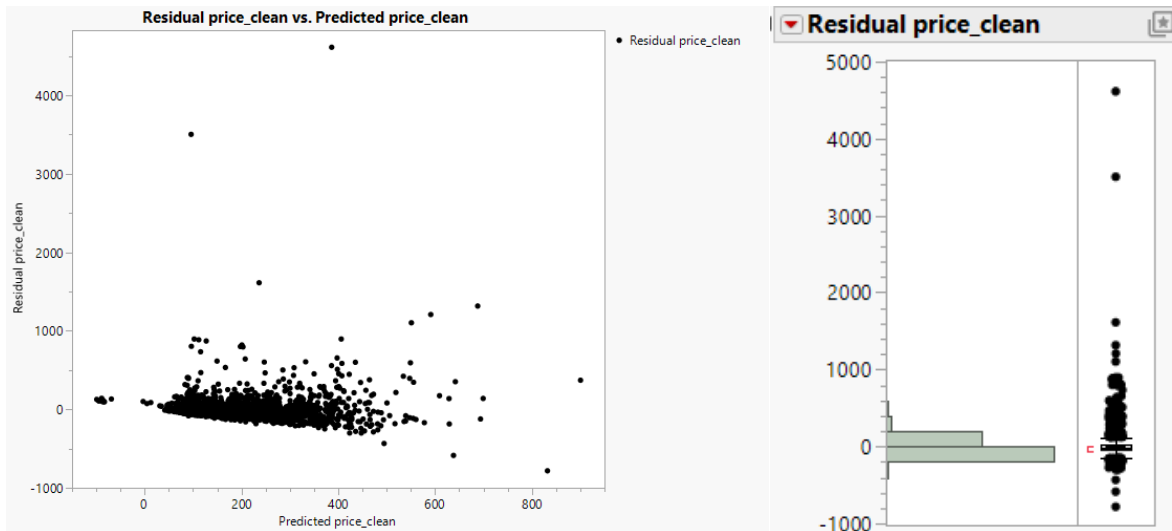
$$H_0 : \beta_i = 0 \text{ where } i = 1 - 9$$

$$H_A : \text{at least one } \beta_i \neq 0 \text{ where } \beta_i = 1 - 9$$

Summary of Fit		Analysis of Variance			
RSquare	0.300486	Source	DF	Sum of Squares	Mean Square
RSquare Adj	0.298808	Model	9	32397211	3599690
Root Mean Square Error	141.778	Error	3752	75418968	20101
Mean of Response	164.6047	C. Total	3761	107816179	
Observations (or Sum Wgts)	3762				Prob > F
					<.0001*

Parameter Estimates				
Term	Estimate	Std Error	t Ratio	Prob> t
Intercept	-117.7643	41.07567	-2.87	0.0042*
accommodates	8.0369271	2.079224	3.87	0.0001*
bathrooms	45.340129	3.524246	12.87	<.0001*
extra_beds	15.068538	2.738617	5.50	<.0001*
bedrooms	26.548312	4.576534	5.80	<.0001*
reviews_per_month	-1.120512	1.122785	-1.00	0.3184
review_scores_rating	26.610833	8.459745	3.15	0.0017*
room_type_Private room	-7.412508	7.187586	-1.03	0.3025
room_type_Shared room	-181.1493	28.20324	-6.42	<.0001*
room_type_Hotel room	17.775356	45.43165	0.39	0.6956

Model 1 explains ~30% of the variation in price given our R^2 value of 0.3005. Our adj R^2 is very similar, indicating that after accounting for the variables, the model maintains a comparable explanatory power. The overall f-test shows that model is statistically significant (p-value < 0.05), providing evidence that at least one of the betas is not equal to zero. We reject the null hypothesis and accept the alternative hypotheses. The parameter estimates show that all the size related variables ($x_1 - x_4$) are significant.



The residuals versus predicted values plot shows clear funnel shape, indicating non-constant variance as the spread of residuals increases with higher predicted prices. The histogram residuals support in showing a strong right skew with several outliers. The residuals are not normally distributed. Model 1 violates key regression assumptions.

Model 2:

Because model 01 produced a relatively low R^2 value I will explore a simpler model focused only on core listing characteristics to see if it will provide similar explanatory power. Model 02 will remove review related variables to allow us to isolate the effect of actual qualities of the property, rather than user reviews.

$$E\{y\} = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_3 + \beta_4x_4 + \beta_7x_7 + \beta_8x_8 + \beta_9x_9$$

Hypothesis:

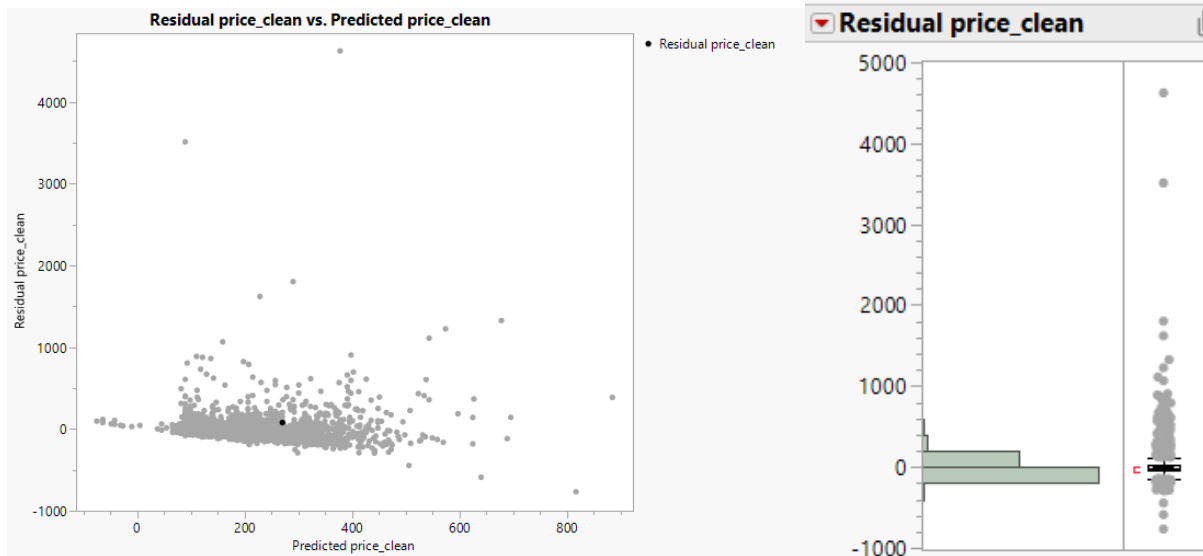
$$H_0 : \beta_i = 0 \text{ where } i = 1 - 4, 7 - 9$$

$$H_A : \text{at least one } \beta_i \neq 0 \text{ where } \beta_i = 1 - 4, 7 - 9$$

Summary of Fit		Analysis of Variance				
RSquare	0.286857					
RSquare Adj	0.285693	Source	DF	Sum of Squares	Mean Square	F Ratio
Root Mean Square Error	141.6854	Model	7	34609248	4944178	246.2884
Mean of Response	163.2445	Error	4286	86040387	20075	Prob > F
Observations (or Sum Wgts)	4294	C. Total	4293	120649635		<.0001*

Parameter Estimates						
Term	Estimate	Std Error	t Ratio	Prob> t	Lower 95%	Upper 95%
Intercept	12.206504	4.839522	2.52	0.0117*	2.7185369	21.694472
accommodates	7.6176637	1.933181	3.94	<.0001*	3.8276285	11.407699
bathrooms	44.06264	3.306475	13.33	<.0001*	37.580238	50.545042
extra_beds	15.458348	2.579914	5.99	<.0001*	10.400381	20.516315
bedrooms	26.646525	4.260281	6.25	<.0001*	18.29417	34.998881
room_type_Private room	-9.112061	6.694734	-1.36	0.1736	-22.23721	4.013083
room_type_Shared room	-155.1413	22.12816	-7.01	<.0001*	-198.5239	-111.7587
room_type_Hotel room	21.645657	45.32236	0.48	0.6330	-67.20964	110.50095

Model 2 explains ~28.7% of the variation in price given our R^2 value of 0.2869. Our adjusted R^2 is very similar, indicating that after accounting for the variables, the model maintains a comparable explanatory power. The overall F-test shows that the model is statistically significant, providing evidence that at least one of the betas is not equal to zero. We reject the null hypothesis in favor of the alternative hypothesis. The parameter estimates show that all the size related variables are significant. We are seeing very similar results to model 1 here.



Again, we are seeing similar results to model 1 here. The residuals versus predicted values plot shows a clear funnel shape, indicating non-constant variance as the spread of residuals increases with higher predicted prices. The histogram of residuals supports this by showing a strong right skew with several outliers. The residuals are not normally distributed. Model 2 violates key regression assumptions.

Model 3:

Because model 1 and model 2 produced very similar results (they are similar models after all) we are unlikely to achieve meaningful improvement in model accuracy through further selection or modification of the IV's themselves. With model 3 the goal is to transform the price data (response variable). The price data is heavily right-skewed and exhibits non-constant variance, so taking the logarithm³ of price helps produce a more normally distributed and centralized response, while also stabilizing variance. This is expected to better satisfy regression assumptions and improve overall model performance.

$$E\{\log(y)\} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + \beta_8 x_8 + \beta_9 x_9$$

Hypothesis:

$$H_0 : \beta_i = 0 \text{ where } i = 1 - 9$$

$$H_A : \text{at least one } \beta_i \neq 0 \text{ where } \beta_i = 1 - 9$$

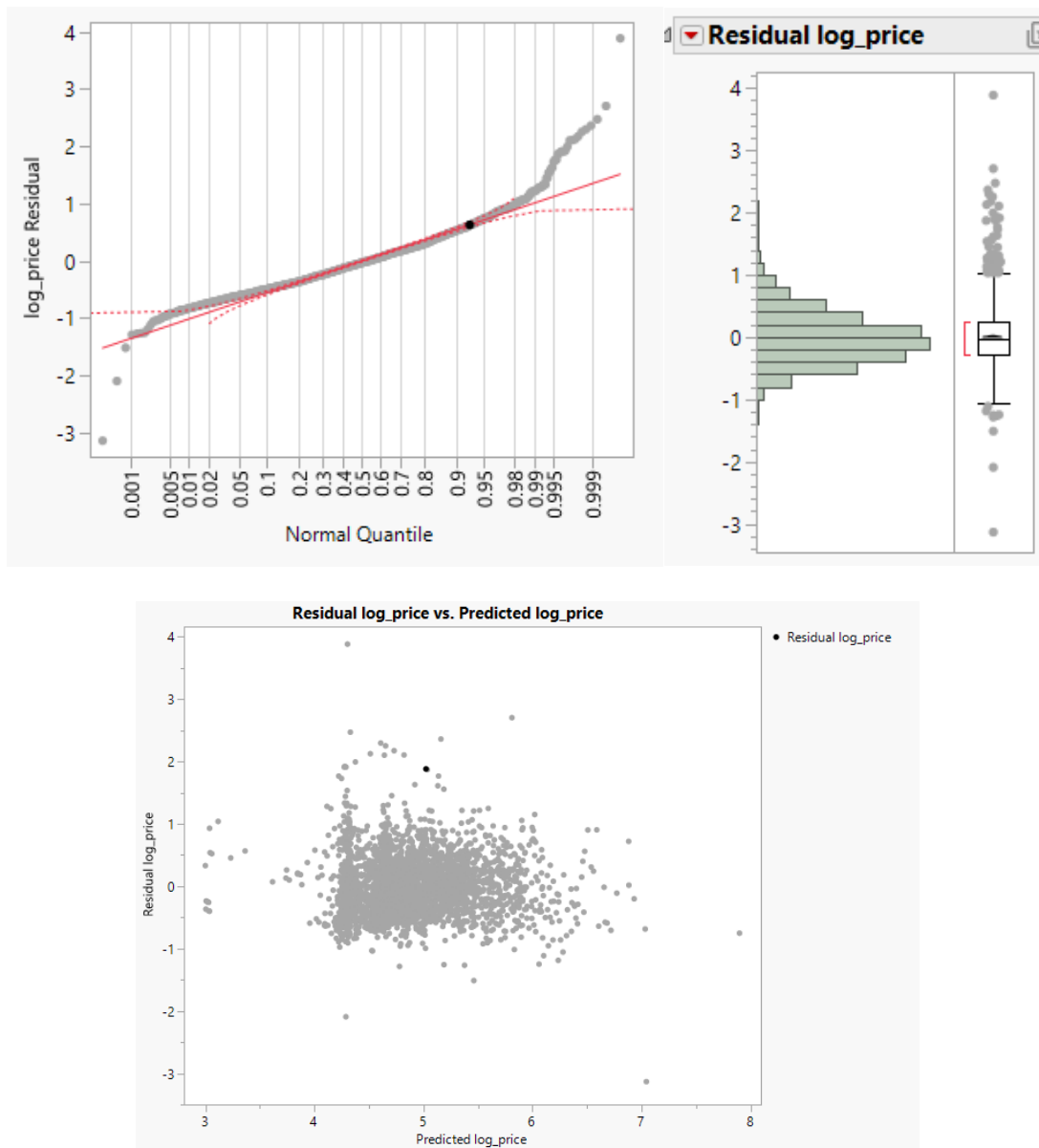
Summary of Fit		Analysis of Variance				
RSquare	0.518422	Source	DF	Sum of Squares	Mean Square	F Ratio
RSquare Adj	0.517267	Model	9	772.7450	85.8606	448.7839
Root Mean Square Error	0.437399	Error	3752	717.8261	0.1913	Prob > F
Mean of Response	4.875957	C. Total	3761	1490.5711		<.0001*
Observations (or Sum Wgts)	3762					

Parameter Estimates						
Term	Estimate	Std Error	t Ratio	Prob> t	Lower 95%	Upper 95%
Intercept	3.4712285	0.126723	27.39	<.0001*	3.2227767	3.7196804
accommodates	0.0545452	0.006415	8.50	<.0001*	0.0419687	0.0671217
bathrooms	0.151153	0.010873	13.90	<.0001*	0.1298361	0.1724699
extra_beds	0.0220271	0.008449	2.61	0.0092*	0.0054622	0.038592
bedrooms	0.1156805	0.014119	8.19	<.0001*	0.0879987	0.1433623
reviews_per_month	0.0110247	0.003464	3.18	0.0015*	0.0042334	0.017816
review_scores_rating	0.1567991	0.026099	6.01	<.0001*	0.1056292	0.2079691
room_type_Private room	-0.327695	0.022174	-14.78	<.0001*	-0.37117	-0.28422
room_type_Shared room	-1.527033	0.08701	-17.55	<.0001*	-1.697624	-1.356442
room_type_Hotel room	0.0327843	0.140161	0.23	0.8151	-0.242015	0.3075839

Model 3 explains ~51.8% of the variation in log(price) given our R^2 value of 0.5184. The adjusted R^2 value is very similar indicating that after accounting for the variables, the model maintains comparable explanatory power. The overall F-test shows that the model is statistically significant, providing evidence that at least one of the betas is not equal to zero. Therefore, we reject the null hypothesis and accept the alternative hypothesis.

It is important to note that in this model due to the transformation of the response variable, the betas are now interpreted in terms of percentage changes in price. The size variables all have significant positive relationships with price, indicating that they drive price up as they themselves increase. For room types, private rooms and shared

rooms are lower in price than baseline (entire home/apt) which intuitively makes sense while hotel rooms are not statistically significantly different in price than baseline.



From the diagnostic graphs above we can now see that the model assumptions are substantially improved compared to previous models. Residuals are more normally distributed and the normal quantile plot residuals more closely resemble the reference line, indicating improved normality. The residuals vs predicted plot no longer exhibits a funnel shape (seen in model 1 and 2) suggesting that non-constant variance has been

reduced. There are still outliers present and visible but it appears the assumption of homoscedasticity has been reasonably satisfied. Model 3 provides a much better fit for the data.

Step History									
Step	Parameter	Action	"Sig Prob"	Seq SS	RSquare	Cp	p	AICc	BIC
1	accommodates	Entered	0.0000	600.9548	0.4032	891.93	2	5257.61	5276.3
2	room_type_Shared room	Entered	0.0000	52.52284	0.4384	619.4	3	5030.68	5055.6
3	bathrooms	Entered	0.0000	55.58671	0.4757	330.85	4	4774.19	4805.34
4	room_type_Private room	Entered	0.0000	42.29134	0.5041	111.8	5	4566.9	4604.28
5	bedrooms	Entered	0.0000	10.28445	0.5110	60.044	6	4516.2	4559.8
6	review_scores_rating	Entered	0.0000	7.838279	0.5162	21.074	7	4477.54	4527.36
7	reviews_per_month	Entered	0.0014	1.952263	0.5175	12.87	8	4469.35	4525.4
8	extra_beds	Entered	0.0091	1.30381	0.5184	8.0547	9	4464.53	4526.8

All Possible Models									
Ordered up to best 4 models up to 8 terms per model.									
Model	Number	RSquare	RMSE	AICc	BIC	Cp			
accommodates	1	0.4032	0.4864	5257.61	5276.30	891.9287	●		
bedrooms	1	0.3874	0.4928	5355.97	5374.66	1015.0988	○		
bathrooms	1	0.2845	0.5326	5939.79	5958.48	1816.3997	○		
room_type_Private room	1	0.1145	0.5925	6741.96	6760.65	3141.2650	○		
accommodates, room_type_Shared room	2	0.4384	0.4719	5030.68	5055.60	619.3975	●		
accommodates, bathrooms	2	0.4320	0.4746	5073.06	5097.98	668.9685	○		
accommodates, bedrooms	2	0.4282	0.4762	5098.49	5123.41	698.9725	○		
accommodates, room_type_Private room	2	0.4252	0.4774	5117.95	5142.87	722.0835	○		
accommodates, bathrooms, room_type_Shared room	3	0.4757	0.4560	4774.19	4805.34	330.8517	●		
accommodates, room_type_Private room, room_type_Shared room	3	0.4628	0.4616	4865.37	4896.52	431.0615	○		
accommodates, bedrooms, room_type_Shared room	3	0.4587	0.4634	4894.53	4925.68	463.6304	○		
accommodates, bathrooms, room_type_Private room	3	0.4571	0.4640	4905.35	4936.50	475.7800	○		
accommodates, bathrooms, room_type_Private room, room_type_Shared room	4	0.5041	0.4436	4566.90	4604.28	111.7994	●		
bathrooms, bedrooms, room_type_Private room, room_type_Shared room	4	0.4888	0.4504	4681.33	4718.70	231.1241	○		
accommodates, bedrooms, room_type_Private room, room_type_Shared room	4	0.4849	0.4521	4709.56	4746.94	261.1345	○		
accommodates, bathrooms, bedrooms, room_type_Shared room	4	0.4821	0.4533	4730.19	4767.56	283.1942	○		
accommodates, bathrooms, bedrooms, room_type_Private room, room_type_Shared room	5	0.5110	0.4405	4516.20	4559.80	60.0437	●		
accommodates, bathrooms, review_scores_rating, room_type_Private room, room_type_Shared room	5	0.5092	0.4413	4529.91	4573.51	73.9514	○		
accommodates, bathrooms, reviews_per_month, room_type_Private room, room_type_Shared room	5	0.5050	0.4432	4561.74	4605.34	106.4408	○		
accommodates, bathrooms, extra_beds, room_type_Private room, room_type_Shared room	5	0.5042	0.4436	4568.23	4611.83	113.1054	○		
accommodates, bathrooms, bedrooms, review_scores_rating, room_type_Private room, room_type_Shared room	6	0.5162	0.4382	4477.54	4527.36	21.0739	●		
accommodates, bathrooms, bedrooms, reviews_per_month, room_type_Private room, room_type_Shared room	6	0.5130	0.4397	4502.93	4552.76	46.6018	○		
accommodates, bathrooms, extra_beds, bedrooms, room_type_Private room, room_type_Shared room	6	0.5118	0.4402	4511.75	4561.58	55.5097	○		
accommodates, bathrooms, bedrooms, room_type_Private room, room_type_Shared room, room_type_Hotel room	6	0.5110	0.4406	4518.16	4567.99	61.9959	○		
accommodates, bathrooms, bedrooms, reviews_per_month, review_scores_rating, room_type_Private room, room_type_Shared room	7	0.5175	0.4377	4469.35	4525.40	12.8696	●		
accommodates, bathrooms, extra_beds, bedrooms, review_scores_rating, room_type_Private room, room_type_Shared room	7	0.5171	0.4379	4472.67	4528.72	16.1924	○		
accommodates, bathrooms, bedrooms, review_scores_rating, room_type_Private room, room_type_Shared room, room_type_Hotel room	7	0.5162	0.4383	4479.47	4535.51	22.9921	○		
accommodates, bathrooms, extra_beds, bedrooms, reviews_per_month, room_type_Private room, room_type_Shared room	7	0.5138	0.4394	4498.51	4554.56	42.1223	○		
accommodates, bathrooms, extra_beds, bedrooms, reviews_per_month, review_scores_rating, room_type_Private room, room_type_Shared room	8	0.5184	0.4373	4464.53	4526.80	8.0547	●		
accommodates, bathrooms, bedrooms, reviews_per_month, review_scores_rating, room_type_Private room, room_type_Shared room, room_type_Hotel room	8	0.5175	0.4377	4471.29	4533.56	14.7969	○		
accommodates, bathrooms, extra_beds, bedrooms, review_scores_rating, room_type_Private room, room_type_Shared room, room_type_Hotel room	8	0.5171	0.4379	4474.62	4536.89	18.1299	○		
accommodates, bathrooms, extra_beds, bedrooms, reviews_per_month, room_type_Private room, room_type_Shared room, room_type_Hotel room	8	0.5138	0.4394	4500.50	4562.76	44.0939	○		

Unfortunately, the stepwise and all possible model outcomes don't offer any big insights that could not have been seen previously. We will use the output from these to determine our final model for use.

Model 04:

Model 04 is meant to piggy back off model 03 using all possible models analysis to construct a simpler model. The goal is to determine whether a simpler model can

maintain similar explanatory power while removing variables that do not contribute meaningfully.

$$E\{\log(y)\} = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_3 + \beta_4x_4 + \beta_5x_5 + \beta_6x_6 + \beta_7x_7 + \beta_8x_8$$

Hypothesis:

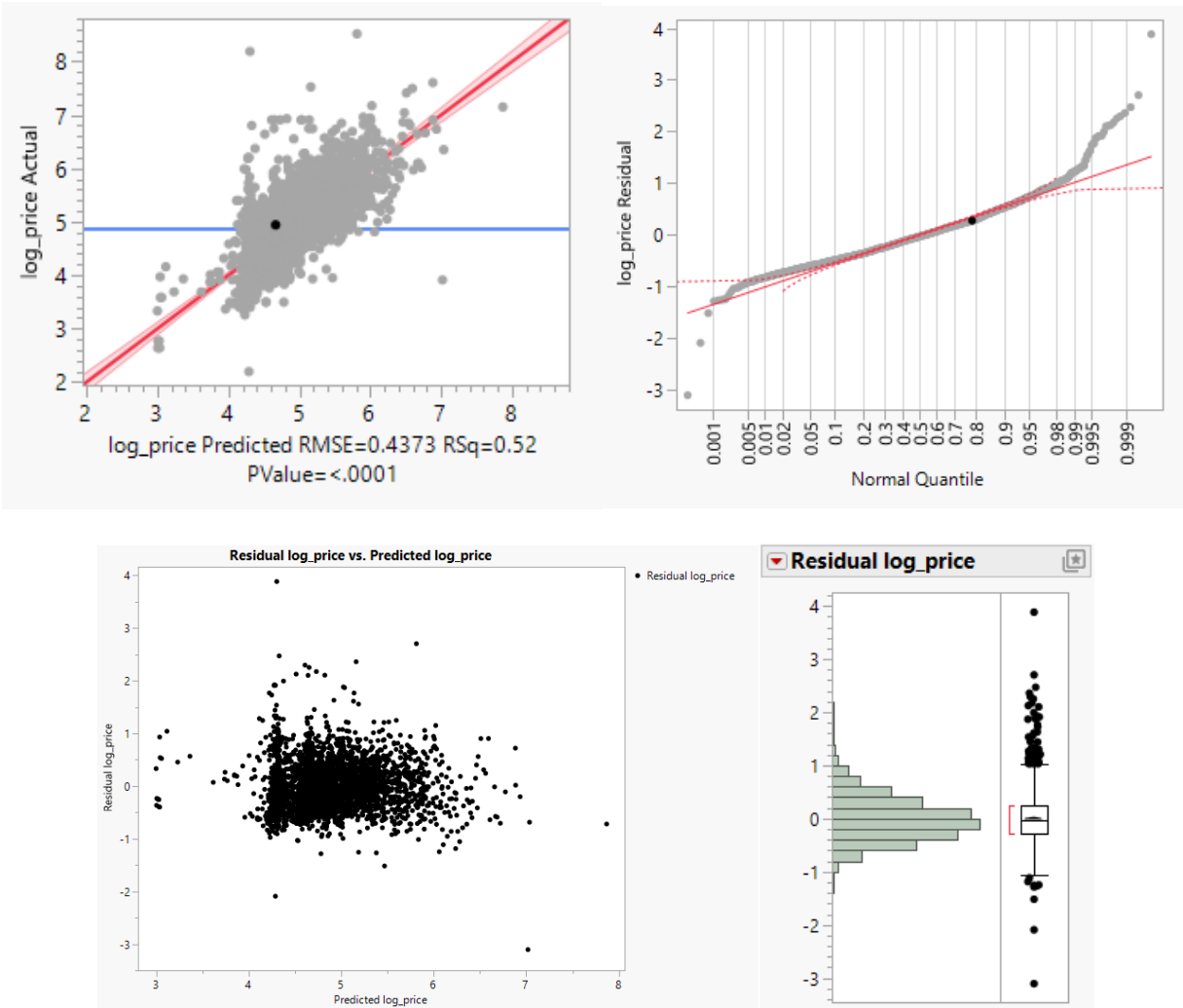
$$H_0 : \beta_i = 0 \text{ where } i = 1 - 8$$

$$H_A : \text{at least one } \beta_i \neq 0 \text{ where } \beta_i = 1 - 8$$

Analysis of Variance					Summary of Fit	
Source	DF	Sum of Squares	Mean Square	F Ratio	RSquare	0.518415
Model	8	772.7345	96.5918	505.0022	RSquare Adj	0.517389
Error	3753	717.8366	0.1913	Prob > F	Root Mean Square Error	0.437344
C. Total	3761	1490.5711			Mean of Response	4.875957
				<.0001*	Observations (or Sum Wgts)	3762

Parameter Estimates						
Term	Estimate	Std Error	t Ratio	Prob> t	Lower 95%	Upper 95%
Intercept	3.4714446	0.126703	27.40	<.0001*	3.2230307	3.7198586
accommodates	0.0544772	0.006407	8.50	<.0001*	0.0419152	0.0670391
bathrooms	0.1515312	0.01075	14.10	<.0001*	0.1304539	0.1726084
extra_beds	0.0220541	0.008447	2.61	0.0091*	0.0054928	0.0386154
bedrooms	0.11563	0.014116	8.19	<.0001*	0.0879549	0.143305
reviews_per_month	0.0110289	0.003463	3.18	0.0015*	0.0042385	0.0178192
review_scores_rating	0.1567326	0.026094	6.01	<.0001*	0.1055721	0.207893
room_type_Private room	-0.32784	0.022163	-14.79	<.0001*	-0.371293	-0.284388
room_type_Shared room	-1.527623	0.086962	-17.57	<.0001*	-1.698121	-1.357125

Model 4 explains ~51.8% of the variation in log(price) and given our adj R² value is nearly identical, we can see that the number of predictors is appropriate. The overall F-test shows that the model is statistically significant, providing evidence that at least one of the betas is not equal to zero. We reject the null hypothesis and accept the alternative hypothesis. Model 4 maintains the accuracy from model 3 but with 1 fewer variable.



The residual diagnostics and findings about model assumptions show little to no change from model 3, further cementing the fact that model 4 is an incremental increase from model 3 without sacrificing performance. Model selection was supported by AIC and BIC values from all possible models analysis. Both log-transformed models offer much better results than models 1 and 2.

RESULTS

Model	Response	Predictors	R ²	Adj R ²	RMSE	F-stat	Prob > F
Model 1	Price	9	0.3005	0.299	141.78	179.08	<0.0001
Model 2	Price	7	0.2869	0.2857	141.69	246.29	<0.0001
Model 3	Log(Price)	9	0.5184	0.5173	0.4374	448.78	<0.0001
Model 4	Log(Price)	8	0.5184	0.5174	0.4373	505.00	<0.0001

Table 01: Model Comparison

The results from the model analysis can be seen from the table above. For detailed outputs and diagnostics on each individual model see the *METHODS AND ANALYSIS* portion of this report.

Exploratory analysis revealed that price is heavily right-skewed with a small number of highly priced listings influencing the distribution. This was always going to cause models with raw price as the predictor to be relatively weak and in violation of key model assumptions. Models 1 and 2 demonstrate this. To address this, the decision was made to transform raw price into $\log(\text{price})$ which reduced skewness and stabilized variance. This would allow models to have more explanatory power and satisfy model assumptions.

A comparison of all 4 models showed that models 1 and 2, using raw price, only accounted for 29-30% of variation and violated regression assumptions. Once models 3 and 4, which used $\log(\text{price})$ were tested, they explained ~51.8% of the variation each.

The final model selected is Model 4. It provides strong explanatory power, satisfies regression assumptions more effectively, and achieves the same level of performance as Model 3 while using fewer predictors. This balance of accuracy and simplicity makes Model 4 the most appropriate choice.

CONCLUSIONS AND DISCUSSION

The purpose of this analysis was to better understand what factors influence Airbnb listing prices and to develop a model that can reasonably explain those differences. The results show that the physical characteristics of a listing play the largest role in determining price. Listings that accommodate more guests, have more

bathrooms and bedrooms, receive better reviews, and offer more privacy tend to have higher prices. This aligns with intuition, as larger and higher-quality listings generally provide more value.

Room type also has a meaningful impact on price. Compared to entire home or apartment listings, private rooms and shared rooms are lower in price. This is not surprising, as Airbnb listings are often used for travel, and privacy tends to be a valuable feature.

One of the most important findings in this analysis is that using raw price did not lead to models that performed well. This was expected, as the data was heavily skewed due to a small number of high-priced listings. By transforming price to a logarithmic scale, the models performed significantly better. This step was critical in developing a model that both fit the data well and satisfied key regression assumptions.

The final model explains over half of the variation in listing prices and provides a strong balance between accuracy and simplicity. Removing variables that did not contribute meaningfully helped simplify the model without reducing performance.

Overall, the analysis shows that price is primarily driven by size, layout, listing type, and reviews. While there is still some variation in price that is not explained by the model, these results directly answer the research question by showing that size, room type, and review-related factors are the primary drivers of Airbnb listing prices.

LIMITATIONS

This analysis is based on observational data, so the relationships identified represent associations rather than direct cause-and-effect relationships. Additionally, some factors that may influence price, such as location specifics, amenities, or seasonal

demand (big one), were not included in the model. While the final model explains a significant portion of the variation in price, there is still unexplained variability.

Additionally, using a log transformation changes how the model is interpreted. The model predicts the logarithm of price rather than price itself, which means results must be converted back to the original scale for practical use. This can introduce slight bias when interpreting predicted prices and may make results less intuitive compared to models using raw prices.

APPENDIX

1. I noticed that beds and bedrooms are highly correlated, since beds are basically determined by bedrooms. I didn't want to use both as separate variables, but I still wanted to capture how much sharing is happening, since when looking at Airbnbs there's usually a mix of people who are okay sharing a room and people who aren't. So I created an "extra beds" variable (beds minus bedrooms) to reflect how many beds are beyond a one-bed-per-room setup, which helps capture that sharing aspect that could impact price.
2. A quick look at outliers showed that it was not apparent that some incorrect information was entered, but that the information was correct. For example there was a listing that had 17 beds also had 7 bedrooms and accommodated 16 which would indicate that the data matches and 17 wasn't linked to a listing with 1 bedroom that accommodates 2. Other examples of outliers followed suit.
3. Because the model predicts $\log(\text{price})$ you need to change the output back to the correct scale. For example if model 4 gives you $E\{\log(\text{price})\} = 5.3$ you would need to do $e^{5.3} = 200.34$ and so \$200.34 would be the expected price.

REFERENCES

<https://insideairbnb.com/>

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